DECISION TREE (CLASSIFICATION)

In [1]:
```
# IMPORTING LIBRARIES
import numpy as np
import pandas as pd
import warnings
warnings.filterwarnings("ignore")
```

In [2]:
```
#LOADING DATASET
from sklearn.datasets import load_wine
data=load_wine()
data
```

Out[2]:
```
{'data': array([[1.423e+01, 1.710e+00, 2.430e+00, ..., 1.040e+00, 3.920e+00,
         1.065e+03],
        [1.320e+01, 1.780e+00, 2.140e+00, ..., 1.050e+00, 3.400e+00,
         1.050e+03],
        [1.316e+01, 2.360e+00, 2.670e+00, ..., 1.030e+00, 3.170e+00,
         1.185e+03],
        ...,
        [1.327e+01, 4.280e+00, 2.260e+00, ..., 5.900e-01, 1.560e+00,
         8.350e+02],
        [1.317e+01, 2.590e+00, 2.370e+00, ..., 6.000e-01, 1.620e+00,
         8.400e+02],
        [1.413e+01, 4.100e+00, 2.740e+00, ..., 6.100e-01, 1.600e+00,
         5.600e+02]], shape=(178, 13)),
 'target': array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2,
        2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
        2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
        2, 2]),
 'frame': None,
 'target_names': array(['class_0', 'class_1', 'class_2'], dtype='<U7'),
 'DESCR': '.. _wine_dataset:\n\nWine recognition dataset\n------------------------
\n\n**Data Set Characteristics:**\n\n:Number of Instances: 178\n:Number of Attribu
tes: 13 numeric, predictive attributes and the class\n:Attribute Information:\n
- Alcohol\n    - Malic acid\n    - Ash\n    - Alcalinity of ash\n    - Magnesium\n
- Total phenols\n    - Flavanoids\n    - Nonflavanoid phenols\n    - Proanthocyani
ns\n    - Color intensity\n    - Hue\n    - OD280/OD315 of diluted wines\n    - Pr
oline\n    - class:\n        - class_0\n        - class_1\n        - class_2\n\n:S
ummary Statistics:\n\n============================== ==== ===== ======= =====\n
Min   Max   Mean     SD\n============================== ==== ===== ======= =====\nA
lcohol:                      11.0  14.8    13.0   0.8\nMalic Acid:
0.74  5.80    2.34  1.12\nAsh:                           1.36  3.23    2.36  0.27\n
Alcalinity of Ash:            10.6  30.0    19.5   3.3\nMagnesium:
70.0 162.0    99.7  14.3\nTotal Phenols:                0.98  3.88    2.29  0.63\n
Flavanoids:                   0.34  5.08    2.03  1.00\nNonflavanoid Phenols:
0.13  0.66    0.36  0.12\nProanthocyanins:              0.41  3.58    1.59  0.57\n
Colour Intensity:              1.3  13.0     5.1   2.3\nHue:
0.48  1.71    0.96  0.23\nOD280/OD315 of diluted wines: 1.27  4.00    2.61  0.71\n
Proline:                       278  1680     746   315\n==========================
=== ==== ===== ======= =====\n\n:Missing Attribute Values: None\n:Class Distributi
on: class_0 (59), class_1 (71), class_2 (48)\n:Creator: R.A. Fisher\n:Donor: Micha
el Marshall (MARSHALL%PLU@io.arc.nasa.gov)\n:Date: July, 1988\n\nThis is a copy of
UCI ML Wine recognition datasets.\nhttps://archive.ics.uci.edu/ml/machine-learning
-databases/wine/wine.data\n\nThe data is the results of a chemical analysis of win
es grown in the same\nregion in Italy by three different cultivators. There are th
irteen different\nmeasurements taken for different constituents found in the three
types of\nwine.\n\nOriginal Owners:\n\nForina, M. et al, PARVUS -\nAn Extendible P
ackage for Data Exploration, Classification and Correlation.\nInstitute of Pharmac
eutical and Food Analysis and Technologies,\nVia Brigata Salerno, 16147 Genoa, Ita
ly.\n\nCitation:\n\nLichman, M. (2013). UCI Machine Learning Repository\n[https://
archive.ics.uci.edu/ml]. Irvine, CA: University of California,\nSchool of Informat
ion and Computer Science.\n\n.. dropdown:: References\n\n    (1) S. Aeberhard, D.
```

```
Coomans and O. de Vel,\n    Comparison of Classifiers in High Dimensional Setting
s,\n    Tech. Rep. no. 92-02, (1992), Dept. of Computer Science and Dept. of\n
Mathematics and Statistics, James Cook University of North Queensland.\n    (Also
submitted to Technometrics).\n\n    The data was used with many others for compari
ng various\n    classifiers. The classes are separable, though only RDA\n    has a
chieved 100% correct classification.\n    (RDA : 100%, QDA 99.4%, LDA 98.9%, 1NN 9
6.1% (z-transformed data))\n    (All results using the leave-one-out technique)\n
\n    (2) S. Aeberhard, D. Coomans and O. de Vel,\n    "THE CLASSIFICATION PERFORM
ANCE OF RDA"\n    Tech. Rep. no. 92-01, (1992), Dept. of Computer Science and Dep
t. of\n    Mathematics and Statistics, James Cook University of North Queenslan
d.\n    (Also submitted to Journal of Chemometrics).\n',
 'feature_names': ['alcohol',
  'malic_acid',
  'ash',
  'alcalinity_of_ash',
  'magnesium',
  'total_phenols',
  'flavanoids',
  'nonflavanoid_phenols',
  'proanthocyanins',
  'color_intensity',
  'hue',
  'od280/od315_of_diluted_wines',
  'proline']}
```

In [3]:
```
#CONVERTING DATA INTO A DATA FRAME
df=pd.DataFrame(np.c_[data.data,data.target],
                columns=[list(data.feature_names)+["target"]])
df
```

Out[3]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nor
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	
...	...	...	...	...	...	...	...	
173	13.71	5.65	2.45	20.5	95.0	1.68	0.61	
174	13.40	3.91	2.48	23.0	102.0	1.80	0.75	
175	13.27	4.28	2.26	20.0	120.0	1.59	0.69	
176	13.17	2.59	2.37	20.0	120.0	1.65	0.68	
177	14.13	4.10	2.74	24.5	96.0	2.05	0.76	

178 rows × 14 columns

In [4]:
```
# DECISION TREE (CLASIFICATION)
# STEP 1: ASSIGNING VARIABLES(INDEPENDENT/DEPENDENT)

# INDEPENDENT VARIABLE
x=df.drop("target",axis=1)

# DEPENDENT VARIABLE
y=df[["target"]]
```

In [15]:
```
# STEP 2:IMPORT TRAIN TEST SPLIT
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test,=train_test_split(x,y,test_size=0.3)
```

In [16]:
```
print("Original size : ",df.shape)
print("x_train size : ",x_train.shape)
print("x_test size : ",x_test.shape)
print("y_train size : ",y_train.shape)
print("y_test size : ",y_test.shape)
```

```
Original size :  (178, 14)
x_train size :  (124, 13)
x_test size :  (54, 13)
y_train size :  (124, 1)
y_test size :  (54, 1)
```

In [17]:
```
# STEP 3: APPLY DECISION TREE (CLASSIFICATION)
from sklearn.tree import DecisionTreeClassifier
classifier=DecisionTreeClassifier()
classifier
```

Out[17]:
```
DecisionTreeClassifier
▶ Parameters
```

In [18]:
```
# STEP 4 :FIT TO DATA
# Use only train(x_train,y_train)
classifier.fit(x_train,y_train)
```

Out[18]:
```
DecisionTreeClassifier
▶ Parameters
```

In [19]:
```
# CHECK ACCURACY
classifier.score(x_test,y_test)
```

Out[19]:
```
0.9074074074074074
```

In [20]:
```
# PREDICTION OF NEW DATA (DATA MUST IN ARRAY FORM)
data=np.array([[14.13,4.10,2.74,24.5,96.0,2.05,0.76,0.56,1.35,9.20,0.61,1.60,560.0]
data
```

Out[20]:
```
array([[ 14.13,   4.1 ,   2.74,  24.5 ,  96.  ,   2.05,   0.76,   0.56,
          1.35,   9.2 ,   0.61,   1.6 , 560.  ]])
```

In [22]:
```
prediction=classifier.predict(data)
prediction
```

Out[22]:
```
array([2.])
```

In [ ]:
```

```